

AAKHYA VERMA

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PROFILE

Analytical problem solver with experience evaluating AI systems, structuring complex data, and building research-focused technology prototypes. Strong foundation in Python, SQL, GenAI, research synthesis, and cross-functional execution, with a focus on turning complex data into useful products.

PROJECTS & EXPERIENCE

VariantLens - AI-Assisted Genomic Research Intelligence Platform ([Link](#))

Jul 2026

Python · Flask · Pandas · Groq LLM · JavaScript · GWAS Catalog · dbSNP

- Built a GenAI product prototype enabling natural-language exploration of 3,447 variant-trait records across 19 trait groups.
- Designed an evidence-retrieval workflow combining GWAS and dbSNP data, genes, statistical associations, and population frequencies into structured insights.
- Integrated a Groq-hosted LLM with citation-based grounding and guardrails; developed Explore, Compare, and Data Lineage features for transparent research workflows.

LLM Evaluation, Health & Nutrition Claims

Jun 2026

Python · Groq API (Llama 3.3 70B) · Prompt Testing · Structured Documentation

- Designed and executed a 33-prompt evaluation across contested science, high-stakes medical claims, and informal/code-mixed inputs, scoring outputs against NIH, Cochrane, ACOG, and AHA sources.
- Identified confidence-calibration and format-compliance failures; built a Python API testing pipeline to run and log prompt batches at scale and translated findings into product-facing QA recommendations.

Dissertation Project, Bioinformatics Centre, SPPU

Aug 2025 - Jun 2026

GROMACS · Molecular Dynamics · Python · Structural Analysis

- Developed 200 ns molecular-dynamics simulations of hemoglobin proteins from 4 species and synthesized 5 structural parameters to compare stability profiles.
- Identified a coupled structural transition at 120–145 ns in Glycera dibranchiata, indicating a potential oxygen-dissociation event.

Bioinformatics Intern (OJT), GeneArche, Pune

May 2025 - Jul 2025

Python · SQL · GWAS Catalog · dbSNP · 1000 Genomes

- Built a variant-phenotype database for 19 health-related traits by consolidating and cross-validating public genomic data; performed QC checks and documented reproducible workflows for team use.

LEADERSHIP & EXTRACURRICULAR

International Academic Conference - Host & Coordinator

Feb 25 - 27, 2026

- Coordinated speakers, delegates, and organizers across a fast-moving 3-day event, resolving logistical ambiguity in real time.

Science Day Exhibition - Budget & Logistics

Feb 2025 - Feb 2026

- Owned budget and logistics for 2 consecutive years for an interactive event reaching 2,000+ school students.

Human Cafe - Member to President

Sep 2021 - May 2024

- Progressed from Member to Vice President to President of a peer mental-health listening club; conducted 300+ one-on-one peer support conversations and led awareness campaigns reaching the wider student body.

PR Coordinator - Enactus, Student Council & Departmental Society

2022 - 2024

- Led sponsorship outreach across 3 organizations, approaching 100+ prospects and closing 30 sponsorships (~30% conversion).

EDUCATION

MSc Bioinformatics, Bioinformatics Centre, Savitribai Phule Pune University

2024 - 2026

CGPA 8.09 · DBT Scholar (Govt. of India) · Qualified IIT JAM, TIFR GS & GAT-B

BSc (Hons.) Biomedical Science, Shaheed Rajguru College of Applied Sciences for Women, University of Delhi 2021 – 2024

CGPA 8.12

SKILLS

Product/AI: GenAI & LLM Evaluation · Prompt Engineering · AI QA/Bulk Testing · Product Prototyping · Research Synthesis · Technical Documentation · Agile/Scrum Fundamentals · Software Testing

Technical: Python · SQL · Pandas · Flask · GROMACS · MS Office Suite · Canva